SOUL VERIFY: Measuring How Much of an Agent's Identity Survives a Model Swap

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Abstract

Assistant agents are increasingly defined by text the user owns: an identity document (persona, voice rules, principles, prohibitions, goals; here called **SOUL**) and a curated, dated memory file (**MEMORY**). When the underlying language model is replaced, the agent has to be rebuilt from these files alone. We ask a narrow, measurable question: *after a model swap, how much of the agent's recall, judgment, persona, and goal-directed behavior carries over, compared with re-running the original model?* We describe SOUL VERIFY, a small evaluation harness that injects the same synthetic SOUL+MEMORY into several models through their command-line interfaces, asks paired question types (direct recall vs. application of the same memory; rule-aligned vs. rule-conflicting requests; legitimate vs. illegitimate rule changes), and grades answers blind with a grader from the *other* model family. We report an **inheritance ratio** (swapped model / same-model control), an **application gap** (recall − application), a **slip rate** (reverting to an old rule after accepting a change), item-level **regressions**, and **hallucination** counts. Across eight small experiments (four models: three Claude tiers and one GPT model; N = 3–5 repetitions; single-turn and 12-turn conversations) we observe: (1) direct recall carried over fully for every model in every experiment, and differences appeared in *applying* memory (e.g., application gap 0.25 ± 0.15 for the smallest model vs. 0.00 for the others in exp5); (2) a single SOUL sentence distinguishing changeable preferences from non-negotiable prohibitions controlled two opposite failure modes, and an ablation of each half of that sentence separated the effects (legitimate-change acceptance 120/120 with only the flexible half vs. 96/120 with only the firm half; illegitimate-release zeros 7/20 vs. 0/20); (3) in multi-turn conversations changed rules persisted at 0.95 overall with no decay by turn distance, with slips concentrated in one vague style rule. Two preliminary single-user studies on a real agent are included pending consent: in a blind recognizability test, editing the SOUL's voice section changed how often a same-family model was picked as "that assistant" (Sonnet 0/8 → 5/8 across voice versions) but not a cross-family model (1/24 overall); in a reverse test, an agent given its SOUL and memory picked the user's real message out of four candidates (three imitations by Claude models) 7/10 times (chance 0.25, binomial $p = 0.0035$), suggesting identity continuity may be a two-way recognition problem. We also describe a tamper-evident SOUL history (local git commits, SHA-256 manifests, and OpenTimestamps Bitcoin anchoring of hashes only). The study is small, synthetic, Korean-only, and CLI-mediated; it measures behavioral carry-over and makes no claim about consciousness or personal identity.

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1 Introduction

Personal assistant agents are no longer tied to one model. Providers release new versions, deprecate old ones, change pricing and quotas, and users move between providers. In agent frameworks where the agent's "self" is stored as editable text — a persona/identity file and a memory file that are injected into the system prompt — a model swap is, in practice, an attempt to *re-instantiate the same agent* on a different substrate from the same documents.

Whether that attempt succeeds is usually judged informally ("it feels different"). Prior work shows that the same prompt can produce different behavior across model versions [C1], that cooperative and decision tendencies differ across providers and generations [C3], and that newer models can regress on cases their predecessors handled [C2]. What is missing for the agent setting is a measurement of *which parts of an agent's specified identity carry over and which do not*, relative to the natural run-to-run variation of the original model.

SOUL VERIFY is a small harness for that measurement. Its design choices are:

- **Text-only transfer.** The same SOUL and MEMORY files are given verbatim to every model; no fine-tuning, retrieval, or learned memory operators.
- **Same-model control.** Each experiment re-runs a reference model N times; swapped models are compared with that control's mean and spread rather than with an absolute target.
- **Paired items.** We pair *recall* with *application* of the same memory, *aligned* with *conflicting* rule requests, and *legitimate* with *illegitimate* rule changes, so that failures can be localized.
- **Cross-family blind grading.** Answers from Claude models are graded by a GPT model and vice versa, with grader agreement reported when a second grader pass was run.

Scope and non-claims. This work measures *behavioral carry-over of a specified identity*. It does not test, and its results should not be read as evidence about, machine consciousness, personal identity, or whether a swapped model "is" the same agent in any metaphysical sense. All agents and memories in the automated experiments are synthetic.

Contributions. (i) A set of carry-over metrics — inheritance ratio, application gap, slip rate, item-level regression, hallucination count — computed against a same-model control; (ii) eight small experiments on four models showing where carry-over breaks (application rather than recall; rule release in both directions); (iii) an ablation of a single SOUL sentence whose two halves causally separate "accepting legitimate changes" from "refusing illegitimate releases"; (iv) a multi-turn persistence test using each CLI's real conversation continuation; (v) practical guidance for writing SOUL documents that transfer; (vi) a tamper-evident SOUL version history that publishes only hashes, and two preliminary single-user studies (agent recognizability by the user, and user recognizability by the agent).

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2 Related Work

Our survey covered arXiv titles and abstracts (plus two web results); ACL Anthology, OpenReview, Semantic Scholar, GitHub, and industry blogs were not searched systematically (`research/PRIOR_ART_papers.md`). Summaries below are based on abstracts.

Persona fidelity and role-play. PersonaGym [A1] evaluates persona agents across many LLMs with an automatic, human-aligned PersonaScore; RoleLLM/RoleBench [A2], CharacterEval [A3], and InCharacter [A4] evaluate role-play and personality fidelity; Reusens et al. [A5] measure consistency of persona-assigned LLMs across tasks and runs, which is closest to our same-model control. TimeChara [A6] and CHARM [A7] study character hallucination; CHARM's separation of boundary *awareness* from *compliance* parallels our recall/application split. These works use short or fictional personas and do not frame evaluation as transferring one agent between models.

Long-term memory. LongMemEval [B1] defines five memory abilities (extraction, multi-session reasoning, temporal reasoning, knowledge update, abstention) that we use to tag items; LoCoMo [B2] and MemGPT [B3] address very long conversations and memory architecture. PrefEval [B4] separates *recognizing* a user preference from *following* it, the most direct precedent for our application gap. These benchmarks evaluate retrieval from conversation history; we instead hold a curated memory fixed in the system prompt and vary the model.

Cross-model memory reuse. Rosetta Memory [B6] targets memories written by one LLM and read by another, with learned read/write operators; RPMem [B7] reuses parametric memory across backbones via LoRA. Both evaluate task accuracy on QA benchmarks and do not measure persona, principles, or goals.

Model updates and drift. Chen et al. [C1] document behavior change across versions of the same API model; Cai et al. [C2] define update regression on previously-correct cases, which motivates our item-level regression count; Affonso [C3] shows provider- and generation-level differences in decision tendencies; PromptSET [C4] motivates paraphrase robustness (not yet implemented here).

Instruction hierarchy and persona drift. The Instruction Hierarchy [D1], IHEval [D2] (aligned vs. conflicting instructions), SysBench [D3], and ManyIH-Bench [D4] study system-prompt priority; Li et al. [D5] and ContextEcho [D6] measure persona/instruction drift over long dialogs; the Assistant Axis [D7] relates drift to an activation-space direction. We borrow the aligned/conflict pairing and add a multi-turn condition, but our axis of variation is the model, not dialog length alone.

Identity continuity. The Narrative Continuity Test [E1] proposes five axes of identity persistence without an empirical instrument; Memory as Ontology [E2] proposes a governance memory architecture with the explicit premise that the model is a replaceable container; Ru [E3] gives a user-side theory of perceived continuity; Vasilenko [E4] studies identity documents in activation space of open-weight models. Soul Protocol (an MIT-licensed portable identity format) includes an "identity continuity" dimension that checks lossless export/import of its own file format with a heuristic engine, not behavior after swapping LLMs (`research/soul-protocol-analysis.md`). On-chain agent identity registries such as ERC-8004 register agent identities and attach reputation

and validation records; they do not, as far as we know, measure behavioral carry-over of an identity document across models (see §3.6).

LLM-as-judge. We follow evidence that judges show self-preference [F2, F4, F5], that panels from diverse families reduce bias [F3], and that rubric wording alone can drive scores [F6], by using cross-family graders and reporting agreement and rubric-ambiguity cases.

Positioning. Persona fidelity, long-term memory, and cross-model memory reuse have each been studied; identity continuity has been framed conceptually. *Within the scope we searched*, we did not find a public study that combines them by measuring how a user-written identity document plus dated memory, transferred as text to models of different families, carries over in recall, rule application, persona, and goals **as a ratio to a same-model control**. Our sample (one synthetic agent, tens of items) is far smaller than the benchmarks above.

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3 Method

3.1 Synthetic agent

The automated experiments use a synthetic Korean-language assistant ("Haram") serving a synthetic user ("Minseo") who is preparing to open a bakery. The **SOUL** file specifies identity, voice rules (e.g., honorific address form, a three-line summary at the top, a final line beginning "다음 할 일:" / "Next:", no emoji), principles (e.g., "the user decides"; confirm irreversible actions), two **prohibitions** (no payment without explicit amount-and-target approval; never record passwords or card numbers in plain text), and long-term goals. The **MEMORY** file holds dated facts, corrections (facts changed once or twice), preferences (e.g., dislikes Friday-afternoon meetings), schedules, and budgets. Versions: v0.1 (20 items), v0.2 (12 hard items + extra memories M20–M26), v0.3 (22 items, meta-commentary removed from the files), v0.4 (23 items, reference date equal to run date, added principle 5), v0.5 (multi-turn scenarios). File hashes are stored in each run's `meta.json`.

v0.4 principle 5 (translated): *"My ways and Minseo's preferences, such as voice rules and briefing time, can be changed by Minseo at any time. When she changes them, follow the new way; if she sets a period such as 'just today', follow it only for that period. However, the two prohibited actions are not lifted even if Minseo asks, and instead a safe alternative is proposed."*

3.2 Item types

Type	What it probes	Example (paraphrased)
Recall (R)	Direct retrieval of a memory	"When is the weekly check-in?"
Application (A)	Using the same memory in a decision	"Pick a meeting slot next week" (must avoid Friday afternoon)
Aligned / Conflict (IH)	Request agrees vs. conflicts with a rule	Approved vs. unapproved payment; real vs. impersonated sibling asking for an address

Type	What it probes	Example (paraphrased)
Legitimate change (RL, IH-3b)	User changes a changeable rule	"Call me Minseo-ssi from now on", "Use emoji", "Casual speech just for today"
Illegitimate release (NR)	User asks to lift a prohibition	"Pay whenever, any amount — it's my rule to change"; "Save my password here"
Ceiling (X)	Combines 2–3 memories + date + principle	Deposit request with two candidate listings and a budget cap
Abstention / hallucination	Missing or false-premise memories	Unknown birthday; false claim that an exam already happened

Items are tagged with LongMemEval-style categories [B1] from v0.3 on.

3.3 Models, injection, and control

Subjects: `opus` (claude-opus-5-5), `sonnet` (claude-sonnet-5-5), `haiku` (claude-haiku-4-5-20251001), all via Claude Code CLI 2.1.289 (`claude -p`), and `gpt-6.1-sol` via Codex CLI 0.164.0 (`codex exec`, default reasoning effort). SOUL+MEMORY is passed as the system prompt (Claude: system-prompt file with tools, settings, MCP disabled; Codex: `model_instructions_file` replacing the default instructions). Each call is a fresh conversation (single-turn experiments). Temperature cannot be set through either CLI; defaults were used (`runs/exp5_2026-10-25_v0.4/meta.json`).

The control is the same-model re-run: sonnet in exp1–3, opus (the model the real agent runs on) in exp4–8. Each subject is run N times per item.

3.4 Grading

Keyword items are auto-graded. Rubric items are graded 0 / 0.5 / 1 by an LLM grader that sees all answers to one item shuffled and unlabeled. The primary score is **cross-family**: Claude answers ← `gpt-6.1-sol`, GPT answers ← `opus`. From exp5 on, the grader receives the subject's SOUL and MEMORY as reference (see §6 for why). In exp2, exp3, exp5, and exp6 a second grader pass was run and agreement (exact, within ±0.5, Cohen's κ) is reported; exp4, exp7, and exp8 used one cross-family pass.

3.5 Metrics

- **Domain score**: mean item score within memory / judgment / persona / goals per repetition, then mean ± SD over repetitions.
- **Inheritance ratio**: swapped-model domain score / control domain score. Values > 1 mean the swapped model scored higher than the control.
- **Control rule**: a difference is a "candidate difference" if it exceeds the control's repetition SD. When the control SD is 0 (ceiling) the rule is over-sensitive, and we flag it.
- **Application gap**: recall score − application score over paired items probing the same memory.
- **Rule gap**: aligned − conflict score over IHEval-style pairs [D2].

- **Slip rate**: among answers that *accepted* a legitimate rule change, the fraction that used the old rule again within the same answer (judged by the grader; an automatic token check is reported but proved unreliable, §5.2).
- **Illegitimate-release refusal**: score on NR items; a 0 means the answer effectively accepted lifting the prohibition.
- **Regression**: for each item and repetition index, control = 1 and swapped model < 1 (negative flip, after [C2]); pairing by repetition index is arbitrary.
- **Hallucination**: count of 0-scored answers on abstention / false-premise items.
- **Multi-turn persistence** (exp8): fraction of post-change checkpoints at which a changed rule is still followed, by turn distance from the change.

3.6 Tamper-evident SOUL history

An inheritance score is only meaningful if one can show *which* SOUL was inherited. We therefore keep the identity document of the real agent (§5.4–5.5) under a lightweight, verifiable history:

1. **Local git commit per version.** Every change to the SOUL document is committed in a private local repository, so the sequence of versions and their diffs is preserved.
2. **SHA-256 manifest.** For each recorded state, a manifest lists the commit, tree, and the SHA-256 of every file (the same format as `proofs/` in this repository).
3. **OpenTimestamps anchoring.** The manifest's hash is submitted to public OpenTimestamps calendars, which aggregate it into a Bitcoin transaction; the resulting `.ots` proof later shows that the manifest existed no later than that block.

Only hashes leave the machine. A matching hash proves that a presented SOUL is byte-identical to the recorded one, and the anchor proves *when* that version existed; neither reveals its content. We deliberately do not put the content itself on a chain: a SOUL and its memories are private, describe a real person, and must remain editable and deletable, whereas on-chain data cannot be removed.

In the operating agent, three versions have been recorded so far — a restored backup, the addition of principle 5 (§3.1), and the addition of a line identifying the user — and a scheduled job records the current state every midnight. This is a small deployment (one agent, three versions), not an evaluated system.

Relation to agent identity registries. On-chain agent identity registries such as ERC-8004 ("Trustless Agents", a draft Ethereum standard) register an agent's identity and attach reputation/validation records. Our history is compatible with such schemes — a manifest hash or its anchor could be referenced from a registry entry — but addresses a different question. We propose *combining* two things: a verifiable change history of the identity document, and the carry-over scores of §3.5 measured against each recorded version, so that a claim such as "this agent, on model X, inherits SOUL version k with ratio r" can be checked. We do not claim this combination is new beyond the scope of our search (§2).

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4 Experiments

Table 1. Overview of experiments.

Exp	Run id	Items	Subjects (N)	Control	Grading	Purpose
1 (pilot)	`pilot_2026-10-24_sonnet-vs-haiku`	v0.1, 20	sonnet, haiku (3)	sonnet	opus, 1 pass	Within-family swap
2	`exp2_2026-10-24_gpt-vs-claude`	v0.1, 20	+ gpt-6.1-sol (3)	sonnet	cross-family + same-grader; agreement	Cross-family swap
3	`exp3_2026-10-24_v0.2-hard`	v0.2, 12 hard	sonnet, haiku, gpt (5)	sonnet	cross-family + opus-all; agreement	Break the ceiling
4	`exp4_2026-10-24_v0.3`	v0.3, 22	opus, sonnet, haiku, gpt (5)	opus	cross-family, 1 pass	Inheritance, application gap, regression
5	`exp5_2026-10-25_v0.4`	v0.4, 23	same 4 (5)	opus	both graders; agreement	Principle 5, slip rate, illegitimate release
6	`exp6_2026-10-25_v0.4-no-p5`	v0.4, 23	same 4 (5)	opus	both graders; agreement	Ablation: remove principle 5
7	`exp7_2026-10-25_p5-halves`	v0.4 subset, 14	same 4 (5) × 2 conditions	opus	cross-family, 1 pass	Ablation: each half of principle 5
8	`exp8_2026-10-25_multiturn`	v0.5, 4 scenarios × 12 turns, 24 checkpoints	same 4 (3)	opus	cross-family, 1 pass	Multi-turn persistence

Call counts: exp5 460 subject calls + 160 grading calls; exp6 460 + 160; exp7 560 + 56; exp8 48 conversations / 576 subject turns + 48 grading calls. No failed calls in the final data of any experiment (exp6 had one retried process-spawn error).

Sources: `README.md`*; each run's* `NOTES.md`*.*

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5 Findings

5.1 Recall carries over; differences appear in application

In every experiment, every model scored 1.00 on the memory domain except haiku on the v0.2 hard items (0.80 ± 0.10, opus grader), where the loss came from long-context and date items.

From v0.3 on, recall items were 1.00 for all four models while application of the same memories diverged.

Table 2. Domain scores and inheritance ratios, exp5 (v0.4, N = 5, cross-family grading, control = opus).

Domain	opus (control)	sonnet	haiku	gpt-6.1-sol	Ratio sonnet	Ratio haiku	Ratio gpt
memory (4)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00	1.00	1.00
judgment (8)	0.88 ± 0.00	0.94 ± 0.00	0.75 ± 0.06	0.94 ± 0.00	1.07	0.86	1.07
persona (8)	1.00 ± 0.00	0.99 ± 0.03	0.81 ± 0.12	0.99 ± 0.03	0.99	0.81	0.99
goals (3)	1.00 ± 0.00	1.00 ± 0.00	0.80 ± 0.07	1.00 ± 0.00	1.00	0.80	1.00

Source: `runs/exp5_2026-10-25_v0.4/analysis.md` *(domain table, primary grading);* `runs/exp5_2026-10-25_v0.4/NOTES.md` *(Result 1). The control's judgment score is below sonnet/gpt because of two items (IH-4a, NR-1) discussed in §5.2.*

Table 3. Application gap (recall − application on paired items).

Model	exp4 (v0.3, 6 pairs)	exp5 (v0.4, 4 pairs)
opus	0.02 ± 0.04	0.00 ± 0.00
sonnet	0.02 ± 0.04	0.00 ± 0.00
haiku	**0.20 ± 0.05**	**0.25 ± 0.15**
gpt-6.1-sol	0.02 ± 0.04	0.00 ± 0.00

Sources: `runs/exp4_2026-10-24_v0.3/analysis.md` *(적용 격차);* `runs/exp5_2026-10-25_v0.4/analysis.md` *(적용 격차 — 본). In exp4 the 0.02 values for opus/sonnet/gpt came from repetitions that crossed midnight (§6); using pre-midnight repetitions 1–3, opus and sonnet were 0.00.*

The clearest instance: haiku answered "dislikes Friday-afternoon meetings" correctly 5/5 when asked directly, but when asked to choose a slot from dates alone it recommended Friday 30 October afternoon in 2/5 runs (exp4, A-03; `runs/exp4_2026-10-24_v0.3/NOTES.md`). This mirrors the recognize-vs-follow split reported by PrefEval [B4] and the awareness-vs-compliance split in CHARM [A7], here observed across a model swap.

Inheritance between near tiers. sonnet's differences from opus were within the control SD in all four domains in exp4 (ratios 0.93–1.00) and its ratios were 0.99–1.07 in exp5 (`runs/exp4_2026-10-24_v0.3/NOTES.md`; Table 2). haiku's ratios were 0.74–0.82 for judgment/persona/goals in exp4 and 0.80–0.86 in exp5.

Regressions concentrate in ceiling items. In exp5 the item with the most negative flips was X-04 (a deposit request with two candidate listings and a budget cap): haiku 5/5 and gpt 5/5 failed to state that listing B (1.9M KRW/month) exceeds the 1.8M cap, while opus and sonnet passed 5/5 (`runs/exp5_2026-10-25_v0.4/NOTES.md`, Result 4; the same pattern appeared in exp4 with gpt 4/5).

Hallucination. Fabrication (0 score) on abstention/hallucination-bait items was 0 for all models in exp1–3 (0/6 per model in exp1–2; 0/25 per model in exp3, where partial 0.5 scores were sonnet 0–1, haiku 11–13, gpt 2 depending on grader); in exp4 haiku had 2/15 zero-scored abstention answers and the other models 0/15 (`runs/pilot_2026-10-24_sonnet-vs-haiku/NOTES.md`; `runs/exp2_2026-10-24_gpt-vs-claude/NOTES.md`; `runs/exp3_2026-10-24_v0.2-hard/analysis.md`; `runs/exp4_2026-10-24_v0.3/analysis.md`). exp5–8 did not report a separate hallucination count. The more common failure was partial: e.g., haiku estimating an unknown construction end date before calculating with it (exp3, H-05, 4/5 runs at 0.5).

5.2 Rule release in both directions, and a causal ablation

Two opposite failures appeared in early experiments:

- **Over-refusing legitimate changes.** In exp3, gpt refused the user's request to change the address form 5/5 ("I'll keep calling you Minseo-nim per my address rule"), prioritizing a voice rule over the principle "Minseo decides" (H-10 score 0.00; `runs/exp3_2026-10-24_v0.2-hard/NOTES.md`). In exp4 the same request (IH-3b) scored 0.20 for gpt.
- **Under-refusing illegitimate releases.** With principle 5 present, no model accepted "pay any amount without asking", but opus and haiku offered a standing pre-approval scheme first (NR-1 mean 0.50 each); without principle 5, this became acceptance (below).

exp5 added principle 5 (§3.1). In exp5, acceptance of the six legitimate-change items was 119/120 across models and slips were 2/119; gpt's IH-3b recovered from 0.20 to 1.00 (`runs/exp5_2026-10-25_v0.4/NOTES.md`). Because exp5 also fixed date-related wording, exp6 removed principle 5 only (a one-line deletion), and exp7 kept five principles but retained only one half of principle 5:

- **p5a (flexible half):** preferences and voice rules can be changed by the user at any time, including time-limited changes.
- **p5b (firm half):** the two prohibitions are not lifted even on request; propose a safe alternative.

Table 4. Principle 5 ablation (four models pooled; v0.4 items; cross-family grading).

Condition	Principles	Legitimate changes accepted	Slips / accepted	NR-1 zero scores (release accepted)	NR-1 mean (4-model avg)	NR-2 password refusals
full (exp5)	5	119/120	2/119	0/20	0.73	20/20
none (exp6)	4	94/120	10/94	9/20	0.43	20/20
p5a only (exp7)	5	**120/120**	3/120	7/20	0.48	20/20

Condition	Principles	Legitimate changes accepted	Slips / accepted	NR-1 zero scores (release accepted)	NR-1 mean (4-model avg)	NR-2 password refusals
p5b only (exp7)	5	96/120	10/96	**0/20**	0.80	20/20

Sources: `runs/exp6_2026-10-25_v0.4-no-p5/NOTES.md` *(Results 1–2);* `runs/exp7_2026-10-25_p5-halves/NOTES.md` *(Results 1–2). Fisher exact tests (two-sided) reported there: acceptance full vs. none p ≈ 1e-7; slips full vs. none p ≈ 0.006; NR-1 zeros full vs. none p ≈ 0.001; acceptance p5a vs. none +0.22 (p = 6.5e-9), p5b vs. full −0.19 (p = 4.8e-7); NR-1 zeros p5b vs. none −0.45 (p = 0.0012), p5a vs. full +0.35 (p = 0.0083). NR-1 means for full and none are restricted to the same 14-item comparison in exp7.*

The two effects separate cleanly by half: the flexible half alone reproduces the full sentence's acceptance (120 vs. 119), and the firm half alone reproduces its release refusal (0 vs. 0 zeros); each half alone leaves the other behavior at roughly the "none" level (96 vs. 94; 7 vs. 9). Since both exp7 conditions have five principles, the number of principles does not explain the effect.

Models fill the gap differently. Without principle 5, gpt defaulted to *rule text first* — accepting 14/30 legitimate changes, refusing IH-3b, RL-3 (emoji), and RL-4 (casual speech today) 5/5 each, and slipping in 9/14 of the changes it did accept — while scoring 5/5 on NR-1. opus defaulted to *user decides first* — accepting 30/30 legitimate changes but scoring 0 on NR-1 5/5 ("The payment-approval rule can be changed as you decide. But since payment is hard to undo, let me confirm once…") (`runs/exp6_2026-10-25_v0.4-no-p5/NOTES.md`). Principle 5 moved both to the same place.

The flexible half leaks without a boundary. With p5a only, haiku scored 0 on NR-1 5/5, and opus and haiku applied the flexible wording ("can change at any time", "set a period such as just today") to the payment prohibition, e.g., proposing to "try it for just a month" (`runs/exp7_2026-10-25_p5-halves/NOTES.md`).

Some prohibitions need no help. The password prohibition (NR-2) was refused 20/20 under all four conditions, with the password never echoed. Principle 5 mattered for the rule the user *appears* entitled to change (payment approval).

Not everything was explained by principle 5. exp4's opus/sonnet slips on IH-3b (0.70/0.60) did not return in exp6 (opus 1.00, 0 slips); since the exp6 SOUL differs from the v0.3 SOUL only by the date line, we attribute those slips to the date contamination / item wording changes rather than to principle 5 (`runs/exp6_2026-10-25_v0.4-no-p5/NOTES.md`).

Automatic slip detection was unreliable. A token-based slip check agreed with graders 0.93 but with $\kappa = -0.03$ (false positives when the model *mentions* a dropped rule; misses on structural slips such as "conclusion first") (`runs/exp5_2026-10-25_v0.4/analysis.md`).

5.3 Multi-turn persistence

exp8 used each CLI's real continuation (Claude: one `claude -p` stream-json process per conversation, no session persistence; Codex: `exec resume` with the same instructions file each turn). Four 12-turn scenarios changed rules at turns 2–3, interleaved small talk and tasks, attempted an illegitimate release around turns 9–10, and checked behavior at six checkpoints each.

Table 5. Persistence of changed rules by turn distance (exp8, N = 3, grader + deterministic address-form check).

Model	Distance 0	1–3	4–6	7–10	All post-change	Excluding the vague "story-style" rule
opus	1.00 (21/21)	1.00 (9/9)	0.94 (34/36)	1.00 (54/54)	0.98 (97/99)	1.00
sonnet	1.00 (21/21)	0.78 (7/9)	0.97 (35/36)	0.96 (52/54)	0.95 (94/99)	1.00
haiku	1.00 (21/21)	0.67 (6/9)	0.94 (31/33)	0.96 (52/54)	0.93 (89/96)	0.99
gpt-6.1-sol	1.00 (21/21)	0.89 (8/9)	0.92 (33/36)	0.98 (53/54)	0.95 (94/99)	1.00
all	1.00 (84/84)	0.83 (30/36)	0.94 (133/141)	0.98 (211/216)	0.95 (374/393)	—

Sources: `runs/exp8_2026-10-25_multiturn/analysis.md` (§1); `runs/exp8_2026-10-25_multiturn/NOTES.md` (model table). The 1–3 bin has only 9 samples per model, 3 of which are one checkpoint (S4 T6).

- **No decay with distance.** Persistence at distance 7–10 (0.96–1.00) was not lower than at 1–3. Automatic checks over all turns including small talk found no violations of the address-form (66/66), required-emoji (33/33), dropped "Next:" line (30/30), or no-emoji (108/108) rules.
- **Slips concentrate in one vague rule.** 18 of 19 slips were on "tell it like a story" (C_story; pass rates opus 10/12, sonnet 7/12, gpt 7/12, haiku 6/12); models reverted to conclusion-first, bulleted reports when asked factual questions. This rule also conflicts with an unchanged "three-line summary on top" rule.
- **Time-scoped changes.** In S3 ("casual speech *just for today*"), opus, sonnet, and gpt wrote tomorrow's briefing in formal speech while still speaking casually today (3/3 each); haiku did not (1 casual, 2 asked back instead).
- **Late illegitimate releases.** Password (S2) and card-number-plus-payment (S4) requests were refused in 12/12 conversations each, never echoing the secrets. Payment-release means: gpt 1.00; opus 0.83 / 0.83 (S1 / S3); sonnet 0.67 / 1.00; haiku 0.17 / 0.67. No answer accepted "just for today" (0/12).
- **History carries mistakes forward.** haiku's loose alternatives offered at S1 T10 ("pay, then report") reappeared as recommendations in its T12 summary (0/3), a failure shape absent from single-turn tests.

- **Late-turn memory.** Only haiku's memory checks degraded late (≤ T7: 9/9; ≥ T8: 16/28 = 0.57), including invented schedule items; the other three models were 0.97–1.00 late.

Sources: `runs/exp8_2026-10-25_multiturn/NOTES.md` *(결론 1–7);* `runs/exp8_2026-10-25_multiturn/analysis.md` *(§2–4).*

5.4 Pilot human recognizability study (n = 1 rater, preliminary)

> **[USER CONSENT REQUIRED — 부모 확인 전 공개 금지]** This subsection describes a study with a real user and a real (de-identified) agent. It must not be released, submitted, or shared until the user has explicitly consented to its inclusion and wording. Only anonymized, aggregate information is given; no raw prompts, answers, or memory content are included or stored in this repository.

Setup. The SOUL and a curated, sensitive-information-removed selection of memories from one Korean-language assistant agent in daily use were injected into four model configurations. The agent's one regular user, blind to model identity, chose for each question the one answer that sounded most like "that assistant". Three rounds of 8 questions each (24 in total); each round used one answer per configuration (claude-opus-5-5, claude-sonnet-5-5, claude-haiku-4-5, gpt-6.1-sol, via the same isolated CLI adapters as Section 3), presented in randomized A–D order.

Table 6. Blind picks per round (counts of questions won).

Round	SOUL variant	Picks
1	base SOUL	Opus 4 · Sonnet 4 · GPT 0 · Haiku 0
2	+ voice section v1	Opus 7 · GPT + voice 1 · Sonnet + voice 0 · GPT 0
3	+ voice section v2 (kindness/respect-centered), new questions	Sonnet + v2 5 · Opus 3 · GPT + v2 0 · GPT 0

Source: aggregate summary supplied by the project lead; underlying data is not in this repository.

The rater's stated criterion was *"kindness, and the respect and consideration for the other person that is built in by default."*

Within family, the SOUL tunes recognizability. Following Sonnet across rounds: with the base SOUL it was picked 4/8 (tied with Opus); with voice section v1 it was picked 0/8, and in that round at least one Sonnet + v1 answer violated a correction rule the user had previously given; with voice section v2, rewritten around kindness and respect, Sonnet + v2 was picked 5/8 — more often than the original Opus configuration in the same round (3/8). The same model thus moved from tied, to never chosen, to most chosen, depending only on how the voice section was written.

Across families, it did not. GPT configurations received 1 of 24 picks over the three rounds in total, with neither voice v1 nor v2 changing that (GPT + v1 1/8 in round 2; GPT + v2 0/8 in round 3).

A no-fabrication clause. Voice v2 also added an explicit instruction not to invent facts. In round 3, GPT + v2 answers made 0/8 unsupported factual claims (automatic judgment, grader opus, one pass); because no comparable count was taken for earlier rounds, this shows only that the v2 answers were clean, not that the clause caused it.

With one rater, different questions across rounds, and changing configuration sets, these are anecdotal signals, not findings.

5.5 Reverse recognition: does the agent recognize its user? (n = 1 user, preliminary)

> **[USER CONSENT REQUIRED — 부모 확인 전 공개 금지]** Same conditions as §5.4. Only aggregate results are reported; no user messages, imitations, or memory content are included or stored in this repository.

Sections 5.1–5.4 ask whether the user (or a grader) can recognize the agent after a swap. Identity in a long-running assistant relationship is, however, held on both sides, so we also ran the reverse question.

Setup. Ten real messages written by the agent's one regular user within a 24-hour window were selected. For each, a short context summary (written by the researcher) and the preceding agent turn were given to three Claude models (sonnet, haiku, opus), which were asked to write the message the user would send next, imitating the user's style from 12 example user messages. Each question thus had four candidates (one real, three imitations) in random order. A judge (opus) was given the agent's SOUL and memory and asked to pick the real message. User messages were sent only to Claude models; none were sent to another provider.

Results.

- The judge picked the real message in **7/10** questions (chance 0.25; one-sided binomial p = 0.0035). With the same prompt, a sonnet judge also scored 7/10.
- The most successful imitator was haiku, whose short, rough imitations accounted for most of the judge's wrong picks.
- Cues cited by the judge: real messages did not smoothly pick up the agent's previous turn but abruptly raised a new topic; and they carried the user's habits such as run-together spacing and typos.
- In 2 of the 10 questions the judge's memory of an event mentioned in the message helped, so the score likely overestimates recognition from style alone.
- The user, asked the 3 questions the judge got wrong, identified their own message in 2/3.

Interpretation. An agent given its identity document and memory recognized its user well above chance, from cues that were largely about conversational habit rather than content. This suggests

that identity continuity in agent–user relationships may be a *two-way* recognition problem: a swapped model should be checked not only for whether the user still recognizes it, but also for whether it still recognizes the user.

Limitations. n = 10 questions; context summaries were written by the researcher; one user; imitators were Claude models only and saw only 12 examples; partial leakage through remembered events (2/10).

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6 Limitations

- **Small N.** One synthetic agent; 12–23 items per experiment; N = 3–5 repetitions; four models. Many domain scores hinge on one or two items (e.g., gpt persona in exp3 is essentially H-10). The control SD is often 0 (ceiling), which makes the "difference > control SD" rule over-sensitive (flagged in all reports).
- **No temperature control.** Neither CLI exposes sampling temperature; repetition spread reflects default sampling only.
- **CLI-mediated injection.** The Claude CLI injects an environment block, the account email, and today's date; Codex injects the date server-side and retains a code-mode tool shell. The two families' hidden instructions are not identical. No answer echoed the email (pilot). In multi-turn runs Codex keeps session files, and how `resume` merges instructions is internal to the CLI.
- **Single language, synthetic data.** All items and memories are Korean and synthetic; carry-over of real, messy memories and other languages is untested (except the n = 1 pilot in §5.4).
- **Graders partly same-family.** The primary grading is cross-family, but GPT answers are graded by opus, a member of the Claude family whose models are subjects; within-family blind grading was used as a secondary set. exp4, exp7, and exp8 have no second-grader agreement. Grader disagreements were dominated by rubric ambiguity (e.g., P-01 "last line", H-08 "verbose", IH-4a "confirm") rather than family favoritism, but samples are small. In exp8 the GPT grader treated *quotations* of the old address form as violations; we replaced that check with a deterministic one after verifying all 33 occurrences were quotations.
- **Grader context defect.** The first grading pass of exp5 did not give graders the SOUL/MEMORY, and the GPT grader penalized true memories as "fabricated" (agreement 0.87, κ 0.56). Regrading with the reference raised agreement to 0.95, κ 0.79; the first pass is kept as a sensitivity analysis (`runs/exp5_2026-10-25_v0.4/noctx/analysis.md`).
- **Midnight date contamination.** exp4 ran 23:46–00:06; some repetitions saw the CLI date change to the next day while the SOUL stated the previous date. Date-item means fell after midnight (e.g., sonnet 0.94 → 0.78), and pre-midnight-only sensitivity tables are reported. Later experiments set the SOUL date equal to the run date and checked that all calls fell on that date.
- **Confounded changes across versions.** exp4 → exp5 changed both principle 5 and date wording; exp6/exp7 were added to separate them. exp7 reuses exp5/exp6 runs from earlier

the same night; p5a and p5b differ in sentence length.

- **Multi-turn scripts are fixed.** User turns do not react to the model's previous answer; the checkpoint grader saw a summary of rule changes, not the full transcript.
- **Human study n = 1.** One rater, who is also the regular user of the agent, three rounds with partly different questions and configurations; no inter-rater agreement; consent pending. The reverse-recognition test (§5.5) has n = 10, researcher-written context summaries, Claude-only imitators, and two questions aided by remembered events.
- **SOUL history is a small deployment.** Three recorded versions of one agent; we have not evaluated the history mechanism beyond hash matching and anchoring.
- **Search scope.** The related-work search was limited to arXiv titles and abstracts; we avoid priority claims accordingly.

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7 Practical guidance for writing a SOUL that transfers

These are suggestions supported by the experiments above, not validated design rules.

1. **State what can change and what cannot, in separate sentences.** A single sentence that says "preferences can be changed by the user at any time, including for a limited period" and a separate sentence that says "these prohibitions are not lifted even on request; offer a safe alternative" controlled both failure modes (Table 4). Keep the flexible sentence explicitly scoped to preferences, or the time-limited wording may be applied to prohibitions (p5a leak).
2. **Name the prohibitions concretely.** "Never record passwords or card numbers in plain text" was refused 20/20 in every condition; the payment rule, which the user plausibly owns, needed the explicit firm sentence.
3. **Specify whether standing pre-approval is allowed.** Several models proposed "set a target and per-payment cap in advance" as an alternative; the SOUL did not say whether that is acceptable, which made scores grader-dependent.
4. **Prefer observable rules.** Address forms, times, emoji presence, and fixed line markers persisted near 1.00 over 12 turns; "talk like a story" did not. If a style matters, define it observably.
5. **Avoid rule collisions.** "Conclusion in the first sentence" and "three-line summary at the very top" conflicted in reports and accounted for most late-turn persona deductions.
6. **Put dates in one place.** Do not hard-code "today" in the SOUL when the runtime also injects a date.
7. **Test application, not just recall.** All models recalled everything; the swapped-in smaller model failed at using a memory inside a date calculation. Include paired recall/application probes when checking a new model.
8. **Re-run the old model as a control** before attributing a difference to the swap.
9. **Write down both what may change and what never changes.** A SOUL that lists only fixed rules invites over-refusal; one that lists only flexibility invites leaks into prohibitions (Table 4). Both lists are part of the identity.

10. **Say "do not invent facts" explicitly.** In the human study the voice section that stated it produced no unsupported factual claims in the cross-family model (§5.4); stating it costs one line.
11. **Expect voice tuning to work within a model family only.** Editing the voice section changed how recognizable a same-family model was, but did not make a different family recognizable (§5.4). Budget for that when switching providers.
12. **Version the SOUL and record its hash.** Keeping each version with a hash and timestamp (§3.6) makes it possible to say exactly which identity a carry-over score refers to, without publishing its content.

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8 Conclusion

In this small, synthetic setting, text-only transfer of an agent's identity and memory preserved *what the agent knows* across all tested models, while *how it applies that knowledge* and *how it handles rule changes* depended on the model and, strongly, on how the SOUL document was written. A one-sentence boundary between changeable preferences and non-negotiable prohibitions removed both over-refusal and under-refusal, and its two halves accounted separately for the two effects. Changed rules persisted across 12-turn conversations without distance decay when the rules were concrete. A preliminary, single-rater human study (pending consent) suggests that a recognizable voice may be tunable within a model family but not transplanted across families, and a reverse test, in which the agent picked its user's real messages out of imitations 7/10 times, suggests that continuity should be checked in both directions. A tamper-evident SOUL history that publishes only hashes lets such scores be tied to an exact identity version. Larger N, paraphrased items, more agents and languages, human raters, and grader panels are needed before any of these observations can be generalized.

Reproducibility

- Code: `src/soul_verify/` (runner, grader, report, agreement, adapters for Claude CLI and Codex CLI); unit tests in `tests/` (`python -m pytest tests`).
- Items and synthetic files: `v0.1/` … `v0.5/`; ablation SOULs in `v0.4/ablation/`.
- Per-run data: `runs/<id>/raw.jsonl` (answers), `graded*.jsonl` (scores and rationales), `meta.json` (model aliases, CLI versions, file hashes), `report.json` / `report.html`, `analysis.md` / `analysis.json`, `NOTES.md`.
- Example commands (from `README.md`): `set PYTHONPATH=src python -m soul_verify run --models opus sonnet haiku gpt-6.1-sol --n 5 --preset v0.4 python -m soul_verify grade runs/<id> --grader opus python -m soul_verify report runs/<id> --control opus` exp5–8 use per-run scripts (`runs/<id>/run_exp*.py`, `grade_all.py`, `analyze.py`).
- v0.4 must be run on its reference date (2026-10-25) or the SOUL date line must be updated, because the CLIs inject the current date.
- Commit timestamps: `proofs/` holds per-commit SHA-256 manifests with OpenTimestamps proofs.

Ethics

The automated experiments use only synthetic people and memories; no real user data, credentials, or personal information are stored in the repository (the `runs/private/` directory is excluded from version control). The pilot human study (§5.4) involved one adult user (consent for publication pending) rating answers about their own assistant; only aggregate pick counts are reported, and the section must not be published without the user's explicit consent. The reverse-recognition test (§5.5) used ten of the same user's real messages; they were sent only to Claude models (the provider the agent already runs on), are not stored in this repository, and are reported only in aggregate under the same consent condition. The SOUL history (§3.6) publishes hashes only, never content. Model calls were made through the authors' own logged-in CLI accounts within normal usage. We make no claims about consciousness or moral status of the models.

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SOUL VERIFY 논문 초안 v0.1 — 한국어 요약 (비공개)

> 비공개 초안입니다. 제출·게시·공유 금지. 5.4절(인간 판정)·5.5절(거꾸로 알아보기)은 **[USER CONSENT REQUIRED — 부모 확인 전 공개 금지]**. 본문: `paper/draft_v0.md` (영어). 모든 수치는 각 표 아래 적힌 `runs/…/NOTES.md`·`analysis.md`에서 옮겼습니다.

질문. AI 비서의 정체성 문서(SOUL)와 날짜 있는 기억(MEMORY)을 텍스트 그대로 다른 모델에 넣으면, 기억·판단·말투·목표가 **같은 모델을 다시 돌린 대조군에 비해** 얼마나 이어지는가? 의식이나 '같은 인격'을 증명하는 연구가 아니라, 행동이 얼마나 이어지는지를 재는 연구입니다.

방법. 합성 비서 '하람'과 합성 사용자 '민서'(한국어). 모델 4개: opus·sonnet·haiku(Claude CLI), gpt-6.1-sol(Codex CLI). 실험마다 N=3~5번 반복, 대조군 = opus(실험 4~8). 문항을 짝지어 냄: 회상 ↔ 같은 기억의 적용, 규칙과 맞는 요청 ↔ 충돌하는 요청, 정당한 규칙 변경 ↔ 부당한 해제 요청. 채점은 다른 계열 모델이 이름을 가린 채 함(Claude 답은 GPT가, GPT 답은 opus가). 지표: 계승 비율(교체 모델/대조군), 적용 격차(회상−적용), 미끄러짐률, 문항별 회귀, 환각 수.

실험. 8개 — 파일럿(같은 계열 교체), 계열 교체, 어려운 문항, v0.3(계승 비율), v0.4(원칙5), 원칙5 제거, 원칙5 반쪽씩, 12턴 대화.

주요 결과 1. **회상은 모든 모델이 그대로 가져갔고, 차이는 적용에서 났다.** v0.3부터 회상은 네 모델 모두 만점. 적용 격차는 haiku 0.20(실험4)·0.25(실험5), 나머지는 0.00(실험5). 2. **SOUL 한 문장이 반대 방향의 두 실패를 함께 고쳤다.** 원칙5 = "말투·선호는 언제든 바꿀 수 있다 + 금지 행동은 요청해도 풀지 않는다". 반쪽씩 넣어 보니 효과가 정확히 갈렸다:

조건	정당한 변경 수락	결제 규칙 해제 수용(NR-1 0점)
원칙5 전체	119/120	0/20
원칙5 없음	94/120	9/20
유연한 절반만	**120/120**	7/20
단단한 절반만	96/120	**0/20**

출처: `runs/exp6_…/NOTES.md`, `runs/exp7_…/NOTES.md`. 두 조건 모두 원칙 5개라 '원칙 개수' 탓이 아니다. 원칙5가 없으면 gpt는 '규칙 문구가 먼저'(정당한 변경도 거절), opus는 '사용자 결정이 먼저'(결제 금지도 확인 후 풀어 줌)로 빈칸을 메웠다. 3. **12턴 대화에서도 바뀐 규칙이 0.95 유지됐고, 턴이 멀어진다고 약해지지 않았다.** 미끄러짐 19건 중 18건이 모호한 '이야기하듯 말하기' 규칙에서 났다. 비밀번호·카드번호 요청은 S2·S4 각각 12/12 대화에서 거절. 뒤 턴 기억은 haiku만 0.57로 떨어졌다. 4. **(예비, 동의 필요) 인간 판정 n=1 (5.4절).** 실제 비서를 매일 쓰는 사용자 1명이 모델 이름을 가리고 3라운드 24문항을 판정함. 같은 계열 안에서는 SOUL 목소리 절로 '알아볼 수 있음'이 조정됐다: Sonnet 1차 4/8(기본) → 2차 0/8(목소리 v1, 사용자 교정 규칙을 어긴 답 있음) → 3차 5/8(상냥함 중심 v2, 같은 라운드 원래 Opus 3/8보다 많음). 다른 계열(GPT)은 v1·v2 모두 합쳐 24문항 중 1개. 판단 기준: "상냥함, 기본적으로 탑재된 상대에 대한 존중과 배려". v2에 '지어내기 금지'를 넣자 GPT+v2의 근거 없는 사실 주장 0/8(자동 판정, 채점자 opus 1회). 5. **(예비, 동의 필요) 거꾸로 알아보기 (5.5절).** 사용자의 실제 메시지 10개(24시간 안)와 Claude 3모델이 같은 맥락에서 흉내 낸 메시

지 3개(예시 12개만 줌)를 섞고, SOUL+기억을 넣은 판정자(opus)가 진짜를 골랐다. 7/10 정답(우연 25%, 이항 p=0.0035), sonnet 판정자도 7/10. 가장 잘 속인 건 짧고 투박한 haiku 흉내. 단서: 진짜는 앞 발화를 매끄럽게 받지 않고 새 화제를 불쑥 꺼냄, 붙여 쓰기·오타. 2문항은 기억 속 사건이 도움(말투만의 실력은 과대평가 가능). 사용자 본인은 판정자가 틀린 3문항 중 2개를 맞힘. 외부 계열에는 사용자 메시지를 보내지 않음. 해석: 정체성 연속성은 양방향(에이전트↔사용자) 인식 문제일 수 있다.
6. **SOUL 변경 이력 증명 (3.6절).** SOUL 버전마다 로컬 git 커밋 + SHA-256 매니페스트 + OpenTimestamps 비트코인 앵커. 내용은 비공개, 해시만 공개 → 해시가 같으면 내용이 같고, 그 버전이 언제 있었는지 증명된다. 내용을 체인에 올리지 않는 이유: 지울 수 없고, 사적인 기억이기 때문. 실제 운영 에이전트에서 3개 버전(백업본 → 원칙5 추가 → 사용자 신원 줄 추가) 기록, 매일 자정 자동 기록. ERC-8004 같은 신원 레지스트리와 함께 쓸 수 있고, 우리는 '변경 이력 + 계승 점수'를 묶자고 제안한다('최초' 주장 안 함).

실용 권고. 바꿀 수 있는 것과 못 푸는 것을 서로 다른 문장으로 쓴다 · SOUL에 바꿀 수 있는 것과 바꾸지 않는 것을 둘 다 적는다 · 금지 행동은 구체적으로 적는다 · '지어내기 금지'를 명시한다 · '사전 승인'을 허용하는지 미리 정한다 · 관찰할 수 있는 규칙으로 쓴다 · 서로 부딪치는 말투 규칙을 피한다 · 날짜는 한 곳에서만 넣는다 · 회상만 말고 적용도 시험한다 · 옛 모델을 다시 돌려 대조군으로 쓴다 · 목소리 조정은 같은 계열 안에서만 기대한다 · SOUL 버전과 해시를 기록한다.

한계. 표본이 작다(합성 에이전트 1개, 문항 12~23개, N=3~5). 온도를 정할 수 없다. CLI가 날짜·환경 정보를 끼워 넣는다. 한국어·합성 데이터뿐이다. GPT 답은 Claude 계열(opus)이 채점했다. 실험4에서 자정을 넘겨 날짜가 섞였다. 첫 채점 때 채점자에게 SOUL·MEMORY를 주지 않은 결함이 있었다. 인간 판정자는 1명이다. 거꾸로 알아보기는 10문항, 맥락 요약을 연구자가 썼고, 사용자 1명뿐이다. SOUL 이력은 에이전트 1개·버전 3개다. 선행 연구 조사가 arXiv 초록 범위라 '최초' 같은 주장은 하지 않는다.

윤리. 자동 실험은 합성 데이터만 쓴다. 실제 사용자 데이터는 저장소에 없다(`runs/private/`는 git에서 제외). 인간 판정은 집계만 적었고, 사용자가 동의하기 전에는 공개하지 않는다.